

JULIAN ESTEBAN RINCÓN RODRÍGUEZ

Computer Science & Artificial Intelligence

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PROFESSIONAL PROFILE

8th-semester Computer Science & AI student at Universidad Sergio Arboleda, specialized in data science, machine learning, and intelligent systems. Hands-on experience building data pipelines, training predictive models (XGBoost, LightGBM, neural networks), cloud infrastructure on AWS, and local AI applications. Proven ability to take projects beyond the academic setting and deliver production-grade, documented work.

EDUCATION

Universidad Sergio Arboleda — Bogotá, Colombia

2021 – Present

B.Sc. Computer Science & Artificial Intelligence | 8th semester of 9

Current courses (8th sem.): Big Data & Data Engineering, Machine Learning, Simulation & Modeling, Neural Networks — Classification (Elective), Engineering Communication, Ethics & Social Responsibility.

FEATURED PROJECTS

Jarvis IronMan — Local AI Voice Assistant

2025 – Present

Python 3.11 · Ollama (llama3.1:8b) · Whisper STT · Piper TTS · Playwright · Telegram API · tkinter

- Complete pipeline Mic → STT → LLM → Actions → TTS running fully offline without cloud services.
- Gmail and Google Calendar integration via OAuth2; browser and desktop automation with Playwright and pywinauto.
- Custom tkinter HUD with wake-word detection; multiprocessing architecture with ZeroMQ for process isolation.
- Active roadmap: vector memory with ChromaDB, Home Assistant integration, openwakeword.

SAVI v2 — Autonomous Real Estate Valuation System

2026

Python · XGBoost · K-Means · MDP · Q-Learning · Deep Q-Network · PyTorch · scikit-learn

- Full RL pipeline: MDP Value Iteration (259 iterations) → Q-Learning tabular (8,000 episodes) → Deep Q-Network (PyTorch, Double DQN, 150 epochs). Final consensus policy across all three agents.
- XGBoost + cluster feature: $R^2 = 0.9609$. Agent decides whether to APPROVE, REVIEW, or REJECT each valuation based on expected financial risk.
- Built on 20,203 properties (2006–2024). IEEE-format technical paper and interactive HTML demo deployed on GitHub Pages.

Big Data Platform — AWS (Chinook / Sakila / Credit)

2026

AWS EC2 · RDS (PostgreSQL / MySQL) · Athena · S3 · Glue · Terraform · React · FastAPI

- Fullstack on Chinook DB: React/Vite frontend + FastAPI backend on separate EC2 instances, PostgreSQL on RDS; 17-section technical document.
- Hive-partitioned external table (credit2) on Athena + S3; resolved case-sensitivity issue in partitions with manual ALTER TABLE.
- NYC Yellow Taxi dataset (94M rows, 12 months) on AWS EMR with PySpark; resolved schema inconsistency across Parquet files using explicit StructType and per-file normalization with unionByName.

Neural Networks & RL Labs

2026

PyTorch · TensorFlow · CUDA · CIFAR-10 · OpenAI Gym

- Reinforcement learning: Q-learning (FrozenLake), REINFORCE (CartPole), and DQN with replay buffer.
- CNN labs: manual convolution, SmallCNN with dropout and weight decay trained on CIFAR-10; professional Word reports.

Ames Housing ML Pipeline

2026

Python · XGBoost · LightGBM · scikit-learn · Pandas · Jupyter Notebook

- Best model: LightGBM $R^2 = 0.75$, MAE $\approx \$23,600$ on $\sim 20,000$ records (2006–2024 dataset).
- Complete pipeline: EDA, feature engineering, K-Means and hierarchical clustering, comparison of 6 algorithms.

Custom Programming Language

2023

Python · ANTLR4 · Tokenization · Interpretation

- Designed and implemented a Python-inspired language with custom tokenizer, lexer, parser, and visitor-pattern execution engine using ANTLR4.

TECHNICAL SKILLS

Languages	Python (intermediate–advanced), Bash Scripting, JavaScript (basic), SQL
ML / AI	scikit-learn, XGBoost, LightGBM, PyTorch, TensorFlow, Whisper, Ollama
Big Data & Cloud	AWS (EC2, RDS, S3, Athena, Glue, EMR), Terraform, Git / GitHub, Docker (basic)
Data & ETL	Pandas, NumPy, Matplotlib, openpyxl, Power BI, Jupyter Notebook, PySpark
Automation	Playwright, pywinauto, Telegram Bot API, Gmail / Calendar OAuth2
Other	ANTLR4, CUDA, NetworkX, ChromaDB, ZeroMQ, tkinter, pytest

ACHIEVEMENTS & RECOGNITION

1st Place — Wisibilízalas International Competition

2019

Universidad Pompeu Fabra · Barcelona, Spain · Shakespeare School, Quito, Ecuador · Senior Category

- 1st place in international competition organized by Universidad Pompeu Fabra, representing Shakespeare School (Ecuador) in the Senior category at age 17.
- Winning project: website highlighting the careers of women in technology and ICT, competing against teams from Latin America and Europe.
- Project available at: sites.google.com/view/warmitics

Independent Project — Jarvis IronMan

Self-directed development outside the academic curriculum

- Fully functional local AI voice assistant built from scratch, without relying on cloud APIs. Personal initiative beyond the academic curriculum.

CERTIFICATIONS

Database Design — Oracle Academy — August 2024

Database Programming with SQL — Oracle Academy — December 2024

LANGUAGES

Spanish: Native • English: B2+ (technical and academic use — documentation, presentations, reports)

Available for internships, freelance projects, or collaborations in AI, Data Science, and Software Development.